

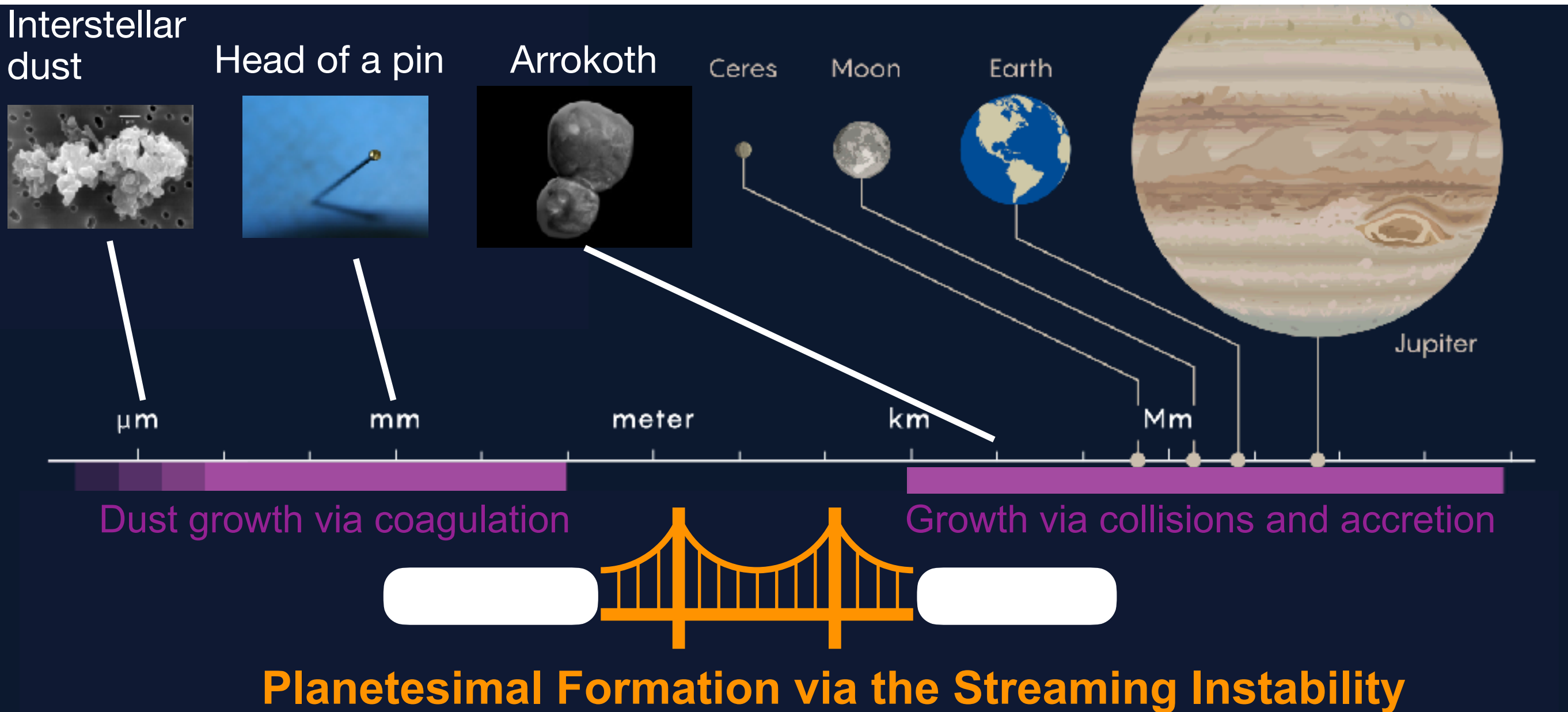
Bridging Unstratified and Stratified Simulations of the Streaming Instability

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The Streaming Instability (SI)

A leading mechanism for **forming planetesimals**



Merrill Sherman/Quanta Magazine

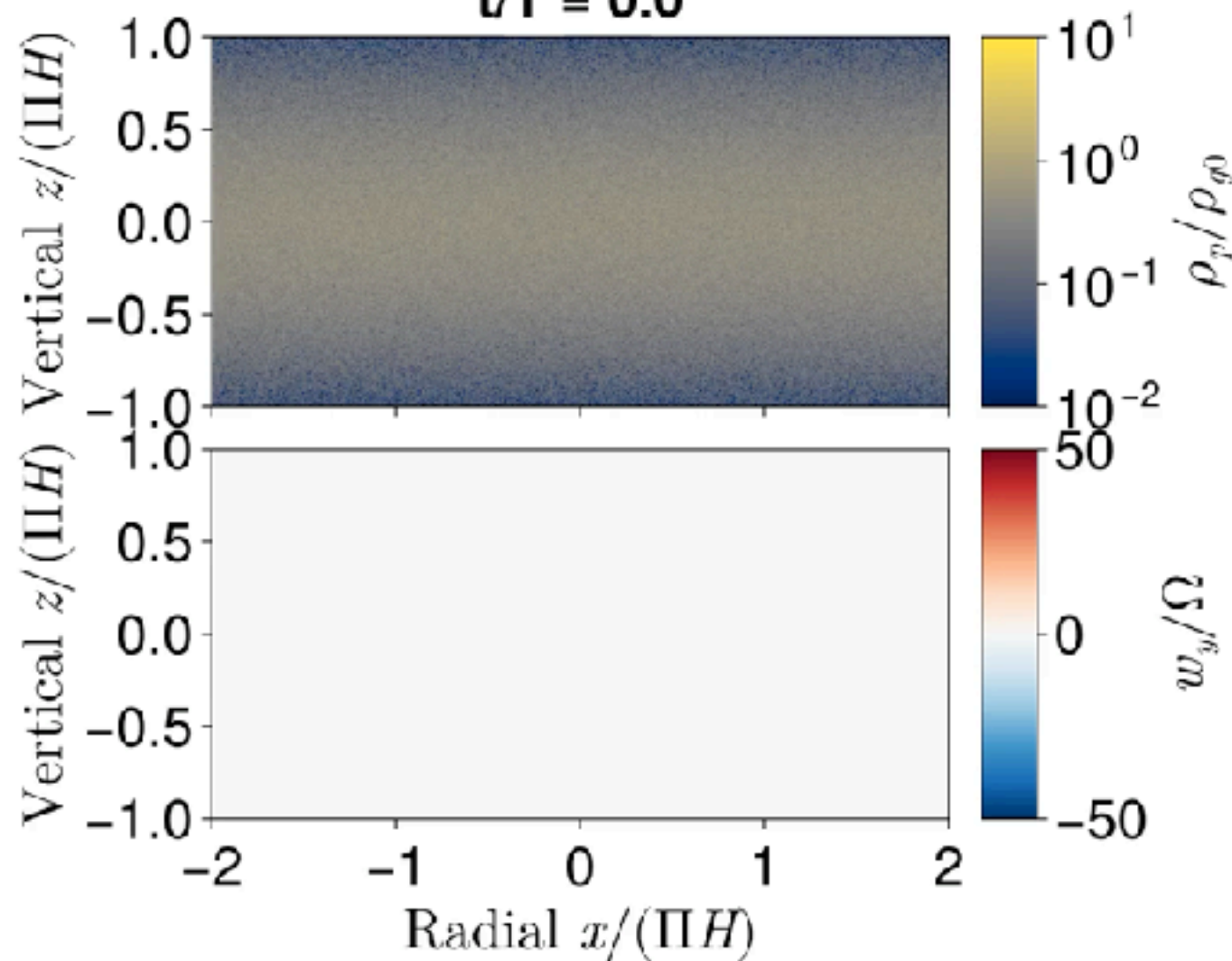
Youdin & Goodman 2005, Youdin & Johansen 2007, Johansen & Youdin 2007; etc.

The Streaming Instability (SI)

A leading mechanism for **driving turbulence near the midplane**

Top: dust density
Bottom: y-component of vorticity

$t/T = 0.0$



Streaming turbulence develops as dust settles, after which the system maintains a self-sustained turbulent state.

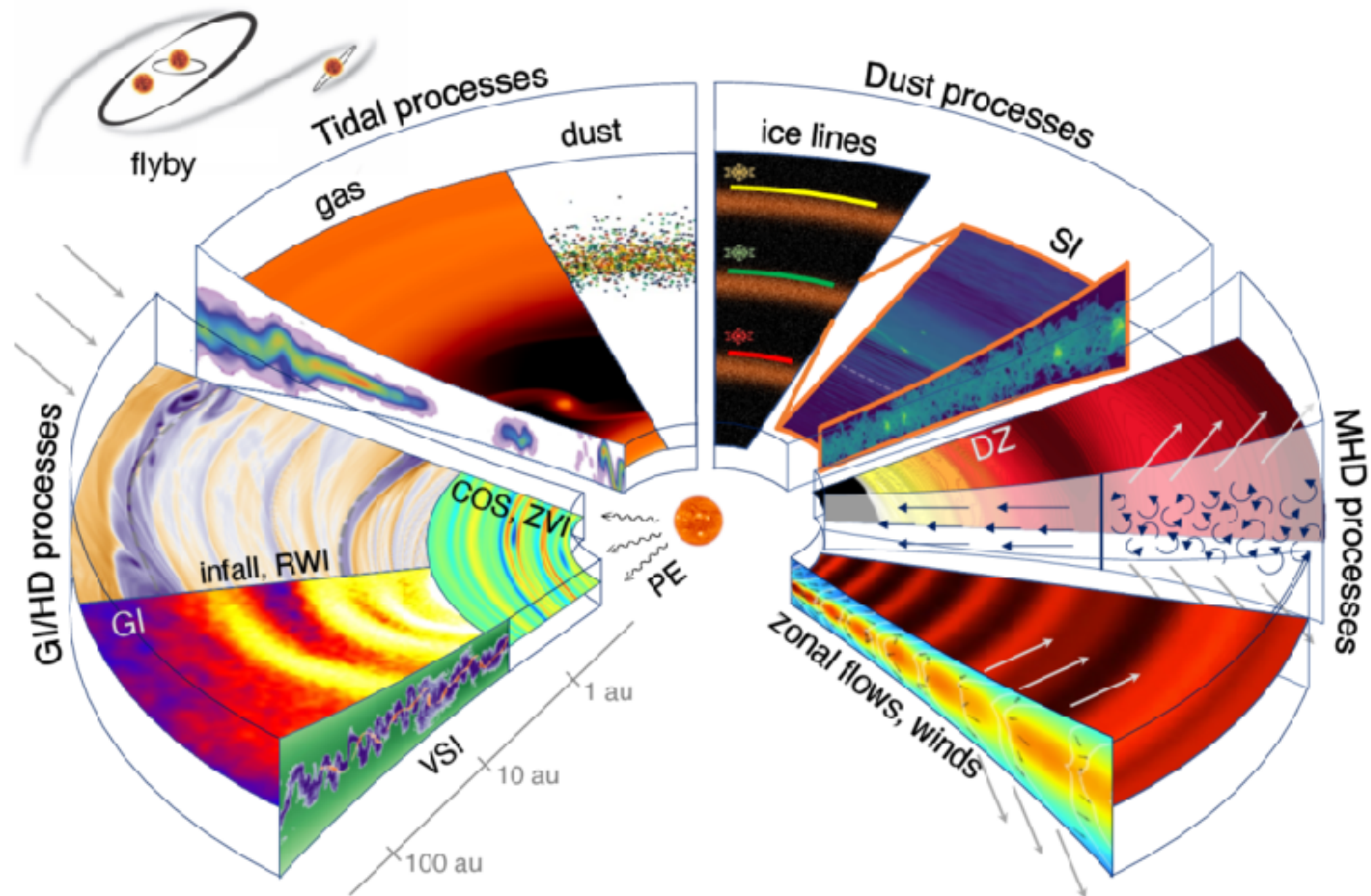
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The Streaming Instability (SI)

Various mechanisms that alter pressure gradients

Radial pressure gradient is both spatially and temporally varying.

The behavior of the SI is affected by pressure gradients.

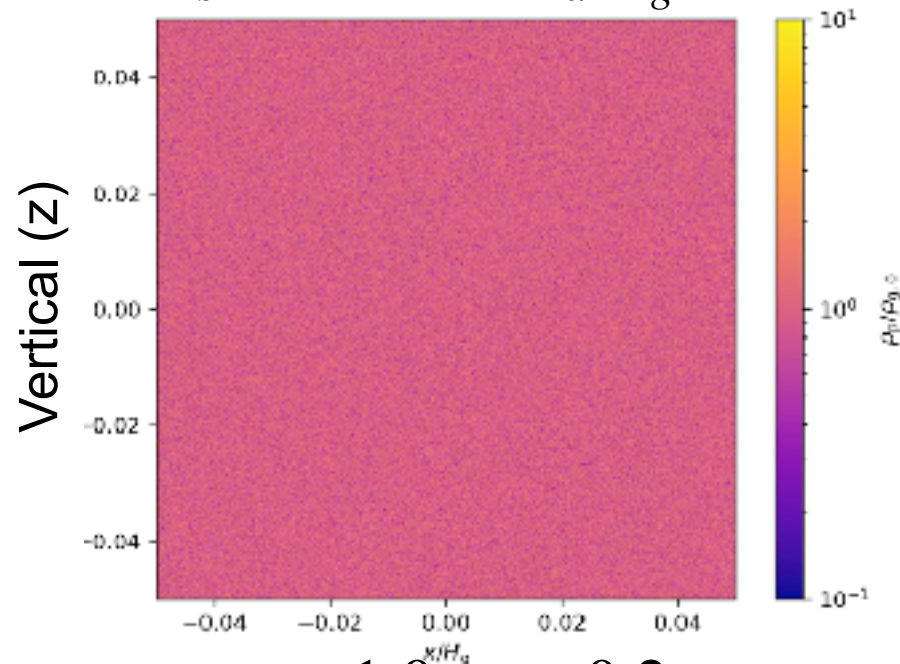


Bae+23

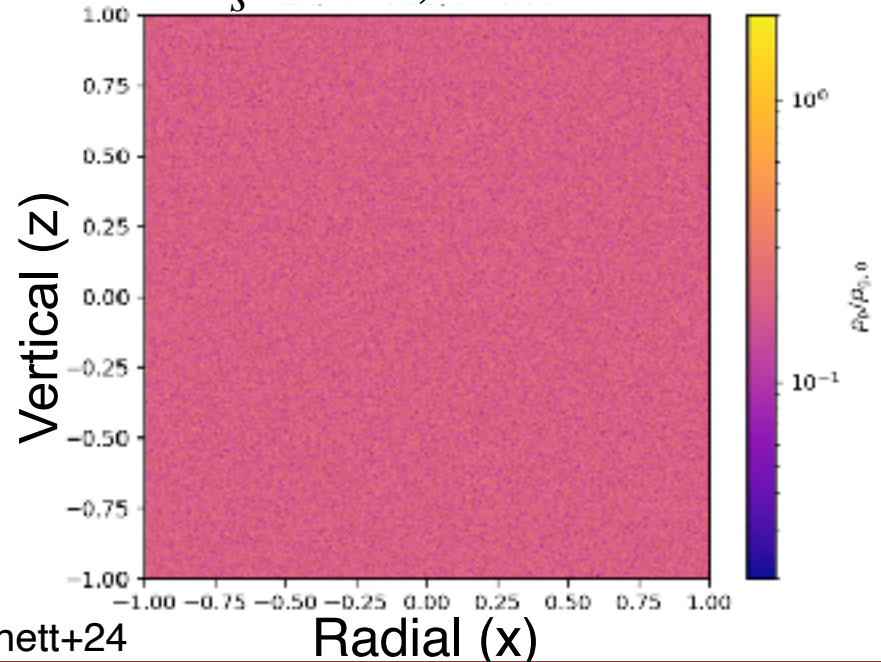
The SI in unstratified simulations

SI-driven dynamics with various pressure gradients in **unstratified** simulations

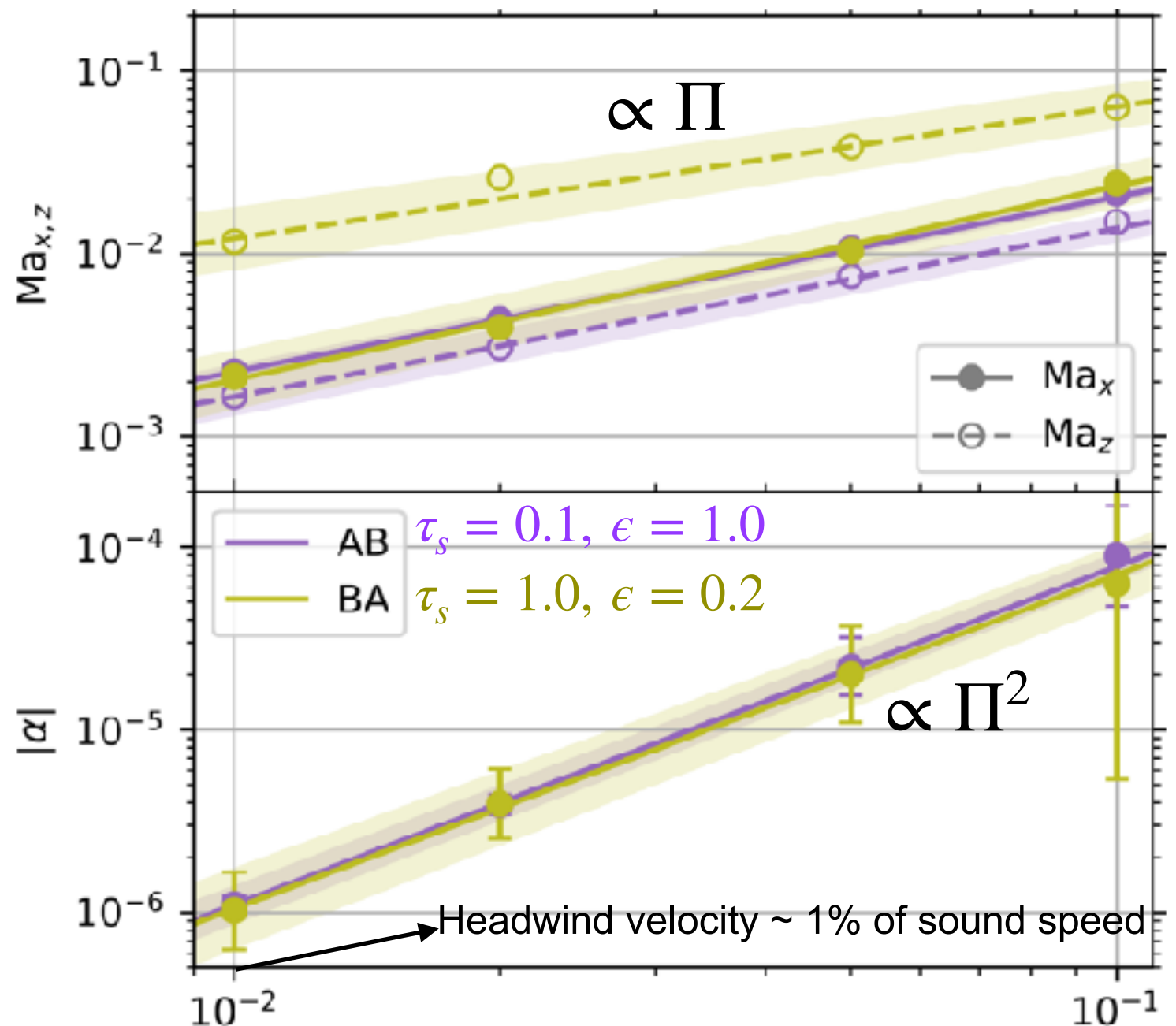
$$\tau_s = 0.1, \epsilon = \rho_d/\rho_g = 1.0$$



$$\tau_s = 1.0, \epsilon = 0.2$$



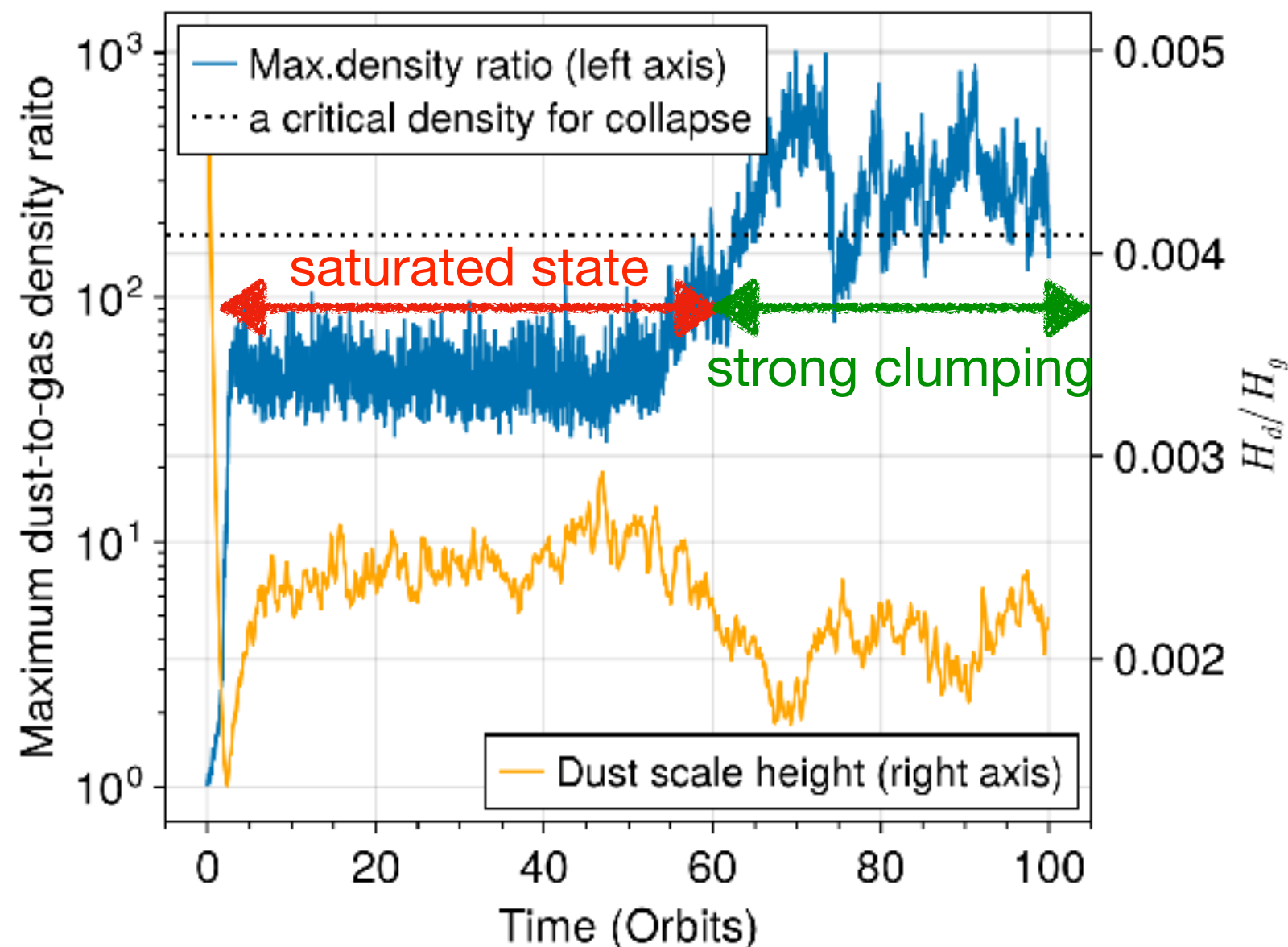
Baronett+24



Π (a dimensionless parameter for pressure gradient; $\propto \nabla P$)

Motivation: The SI-driven dynamics in *stratified* simulations

The saturated state is a **precursor** to strong clumping and planetesimal formation.



1. Turbulence and dust sedimentation play a crucial role in triggering strong clumping.

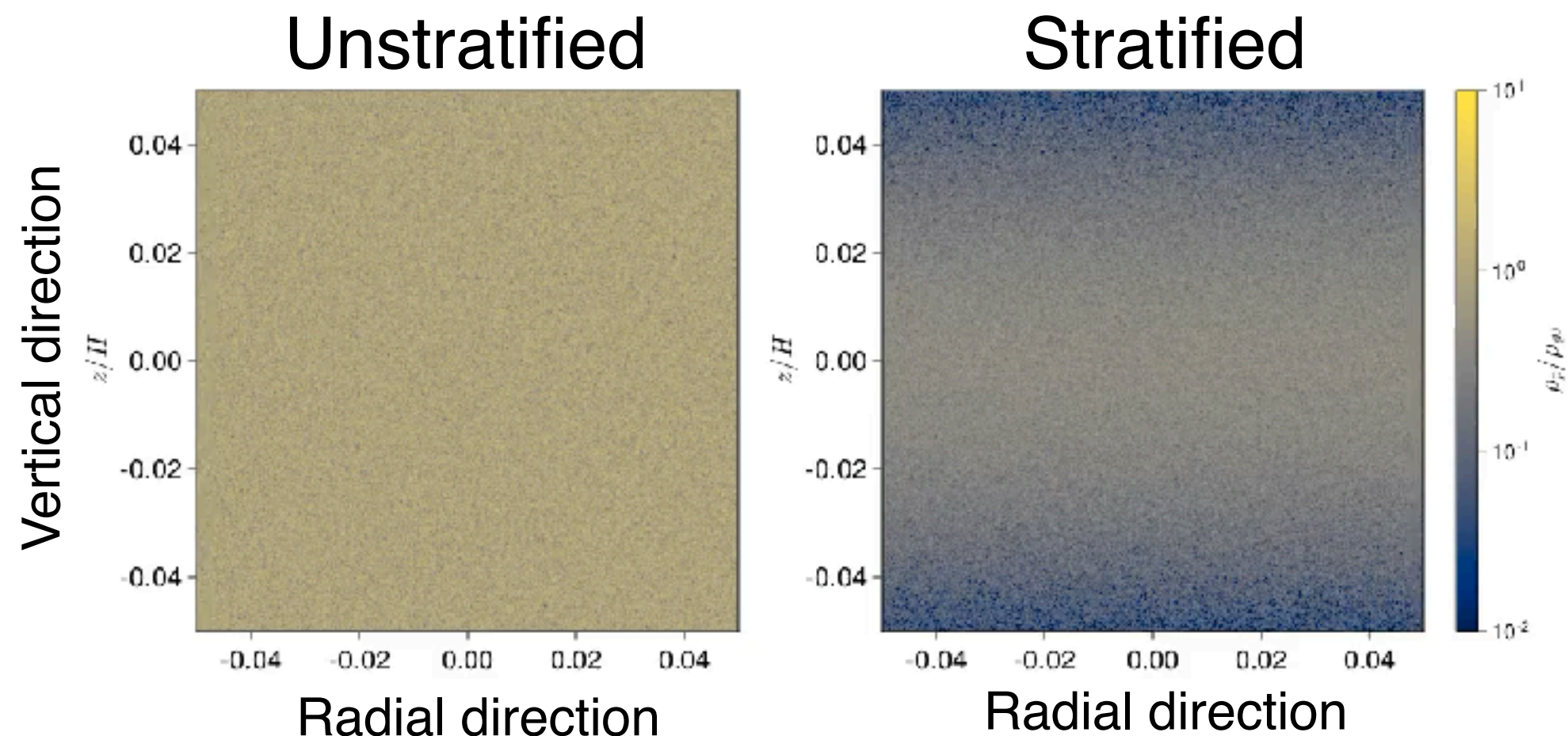
2. How does the saturated state determine strong clumping?

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Motivation: Comparing unstratified and stratified simulations

Unstratified simulations are regarded as an approximation for the disk midplane.

-> How well can unstratified simulations represent dynamics around the midplane of stratified simulations?



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Numerical method

2D axisymmetric, local simulations of stratified disks with $\tau_s = 0.1$

Four simulations considered

Table 1
Model Parameters

$$Z = \Sigma_d / \Sigma_g$$

Z (1)	Π (2)	$L_x \times L_z / (\Pi H)^2$ (3)	$N_{\Pi H}$ (4)	n_p (5)
0.0024	0.01	4×8	256	1
0.0048	0.02	4×8	256	1
0.012	0.05	4×8	256	1
0.024	0.1	4×8	256	1

$$Z/\Pi = 0.24$$

Domain sizes are
in units of ΠH

of grid cells
per ΠH

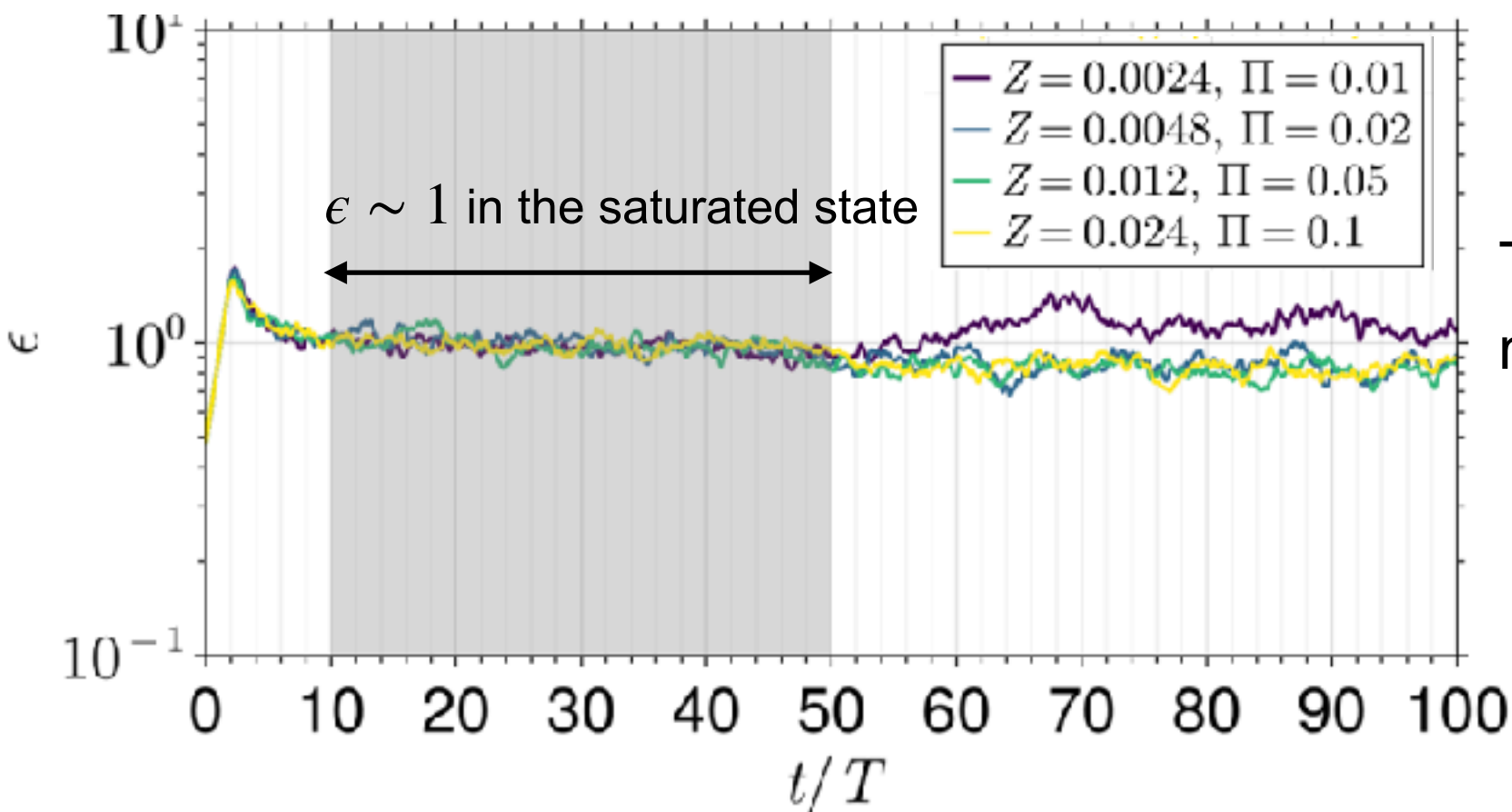
One particle
per cell on
average

Numerical method

Results from our stratified simulations are compared to those in unstratified simulations in Baronett et al. 2024 (B24).

Model AB in B24: $\Pi = 0.01, 0.02, 0.05, 0.1$ at $(\tau_s, \epsilon) = (0.1, 1.0)$

Time evolution of ϵ in **stratified** simulations



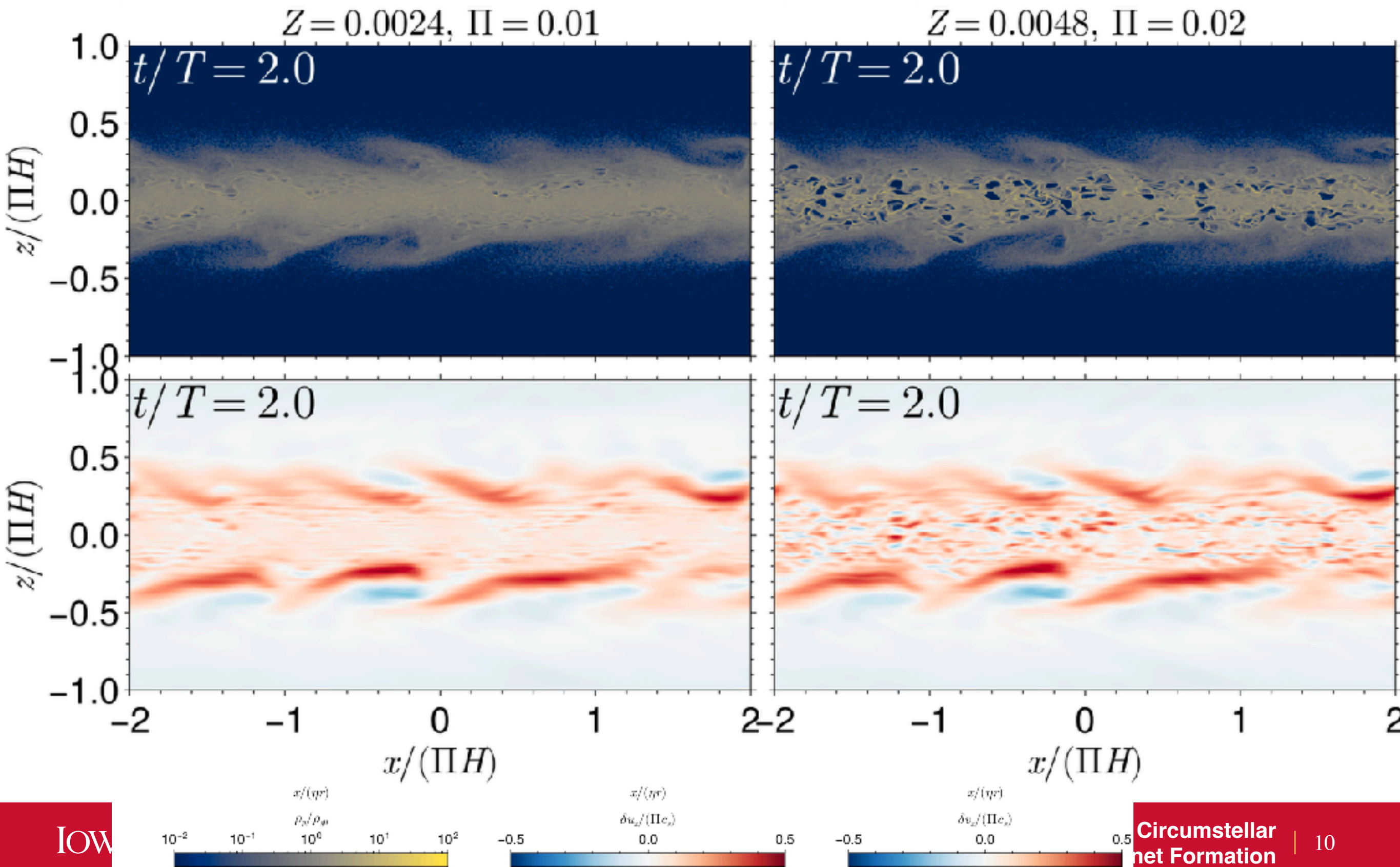
The value of Z/Π is chosen to match ϵ value in Model AB in B24

Results: Sedimentation phase

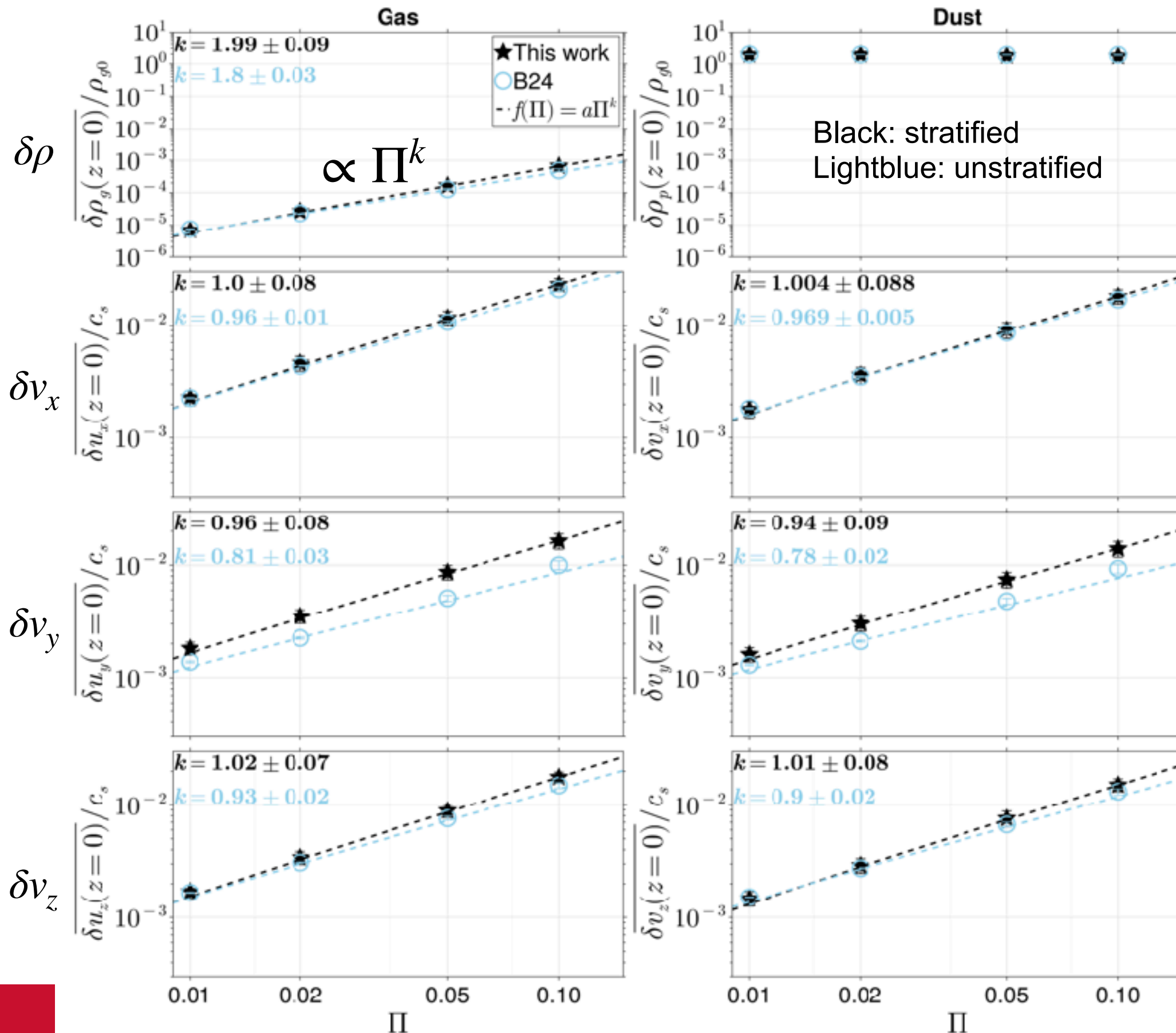
Dust density

Radial velocity of gas

Radial velocity of dust



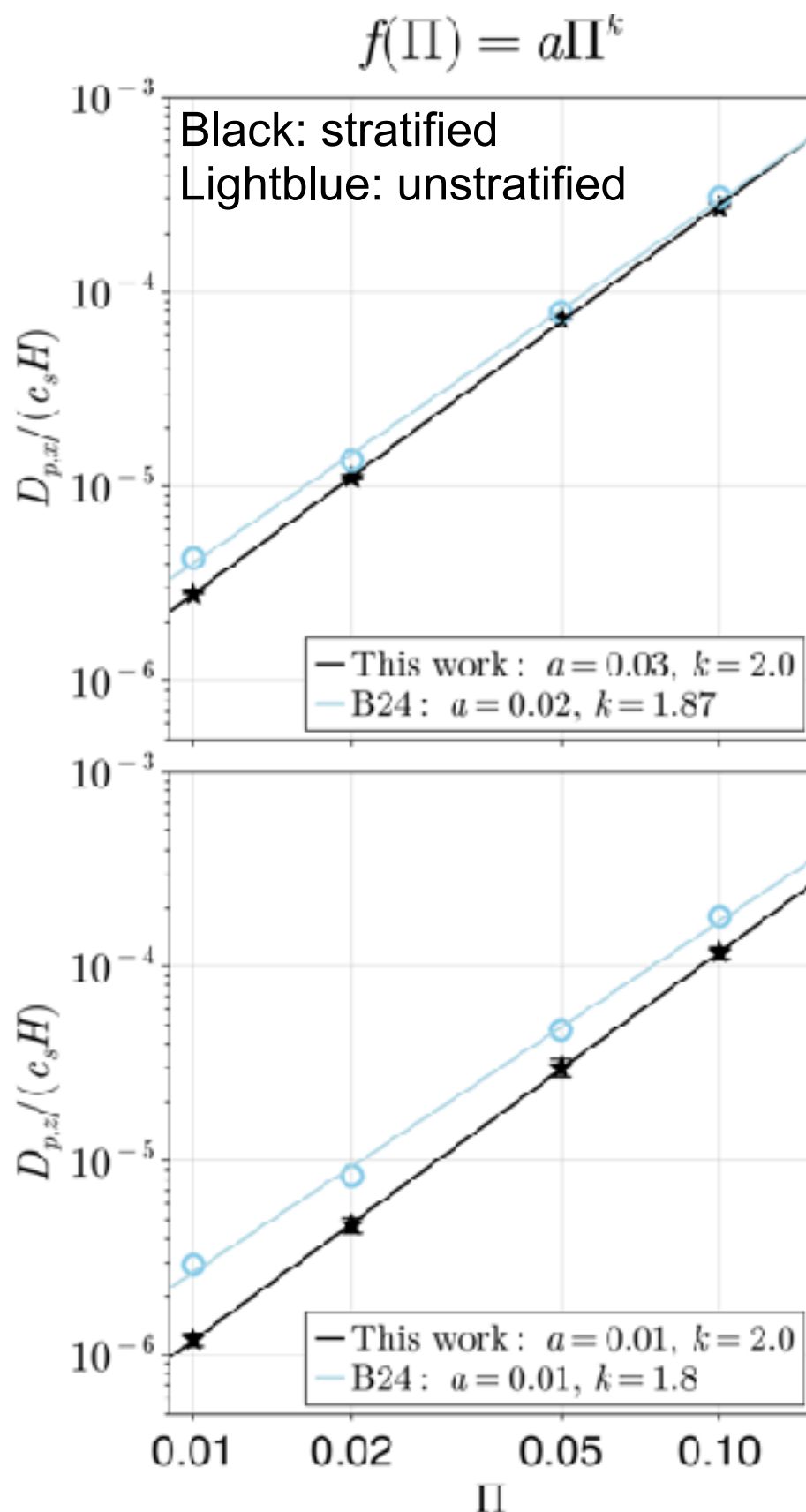
Results: Time-averaged dispersions at the midplane in the saturated state



Dispersion quantities
**are remarkably
 similar** between
 unstratified and
 stratified simulations

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Results: Dust diffusion (radial and vertical)

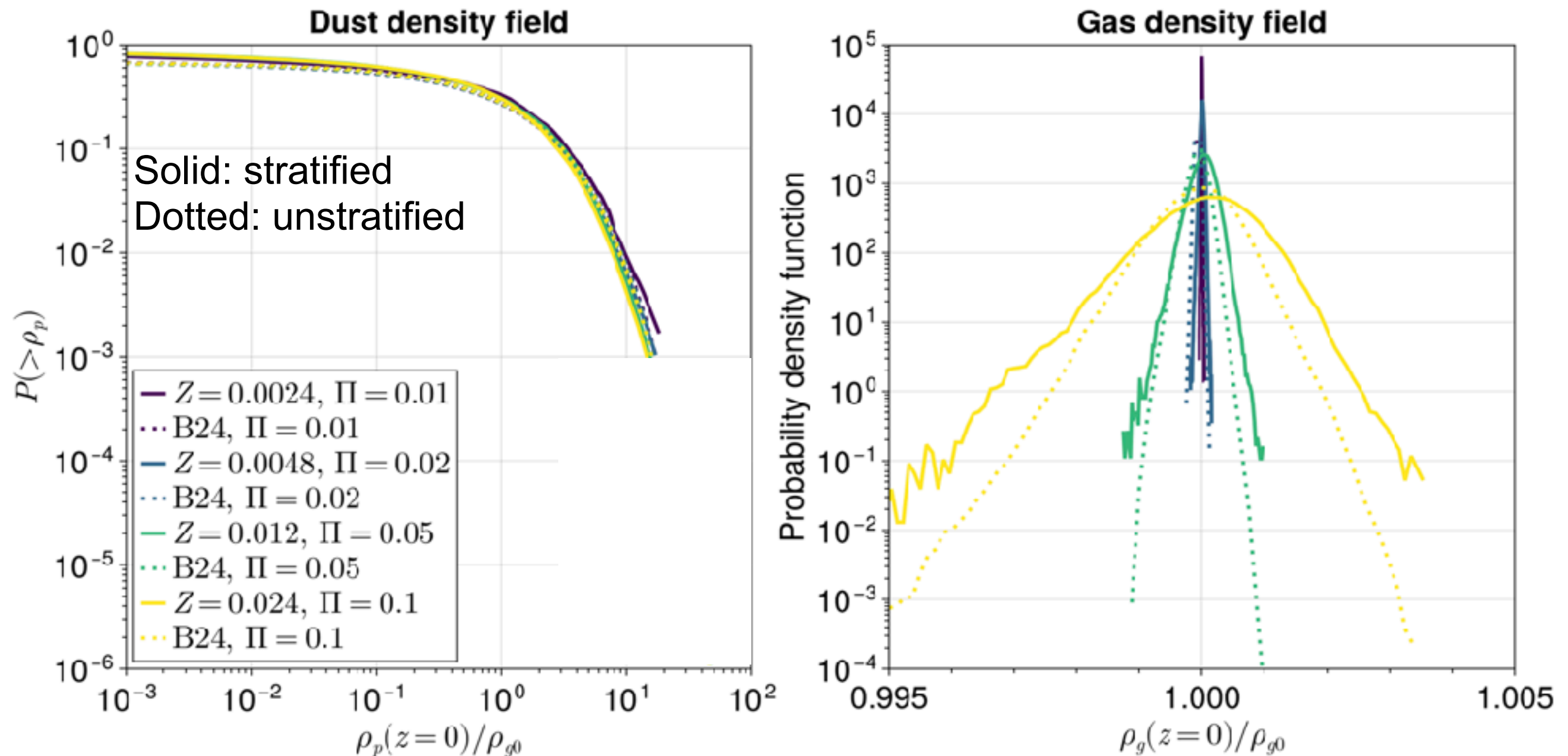


Radial diffusion is similar -> particles undergo a random walk in both unstratified and stratified simulations.

Vertical diffusion is weaker in stratified simulations -> vertical gravity prevents particles from diffusing freely in z .

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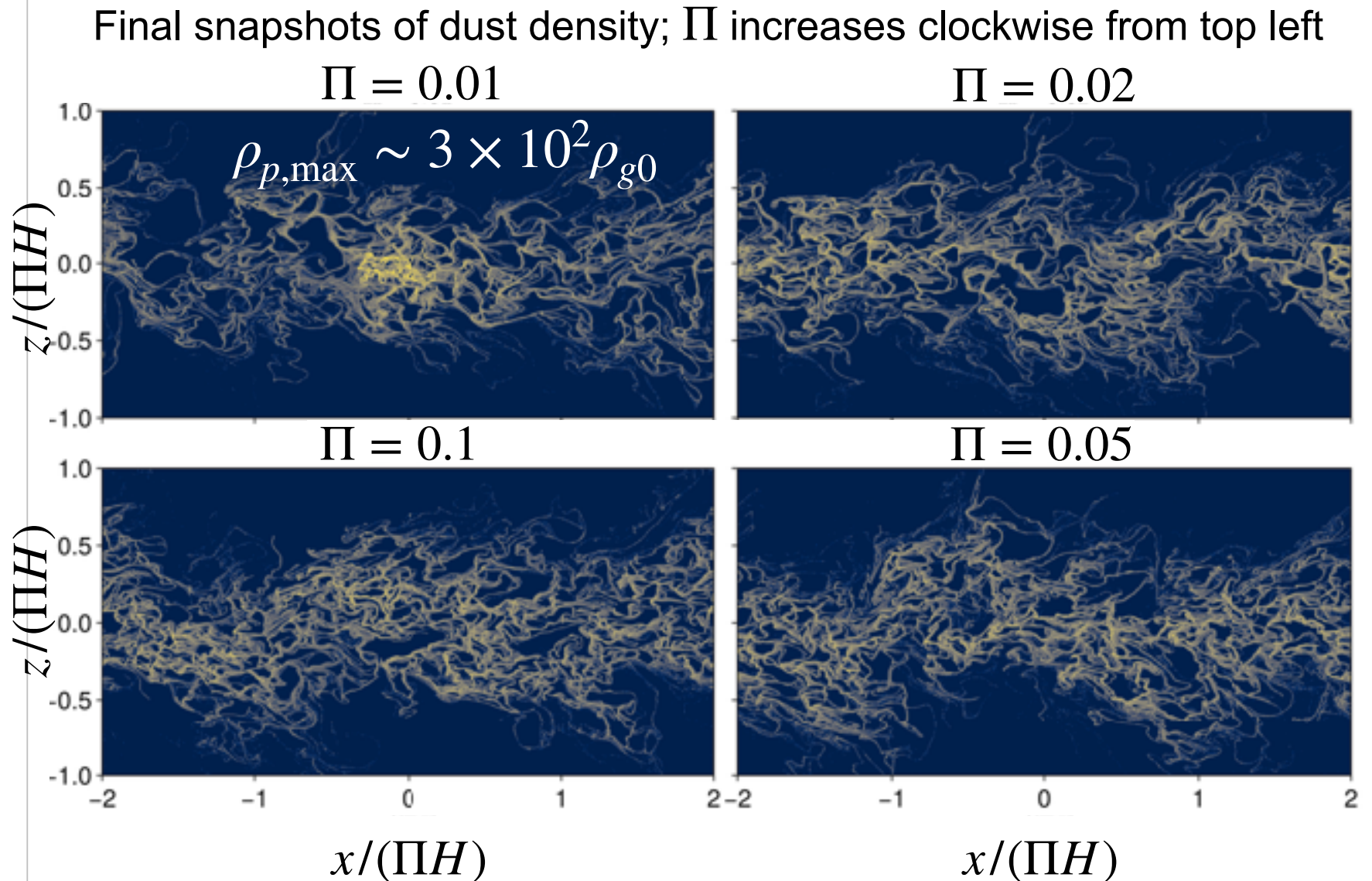
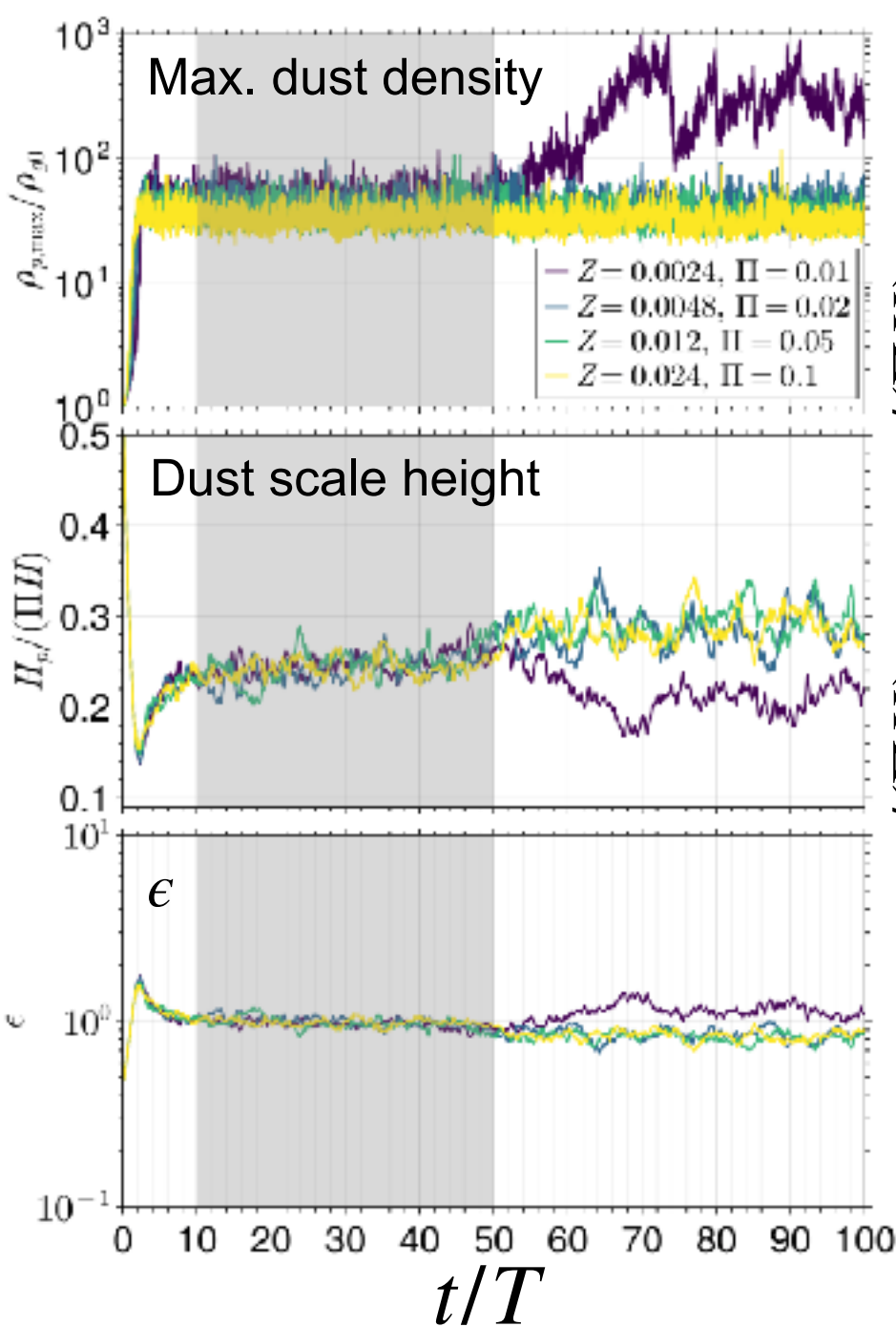
Results: Density distributions at the midplane



Both gas and dust densities exhibit similar distributions between unstratified and stratified models

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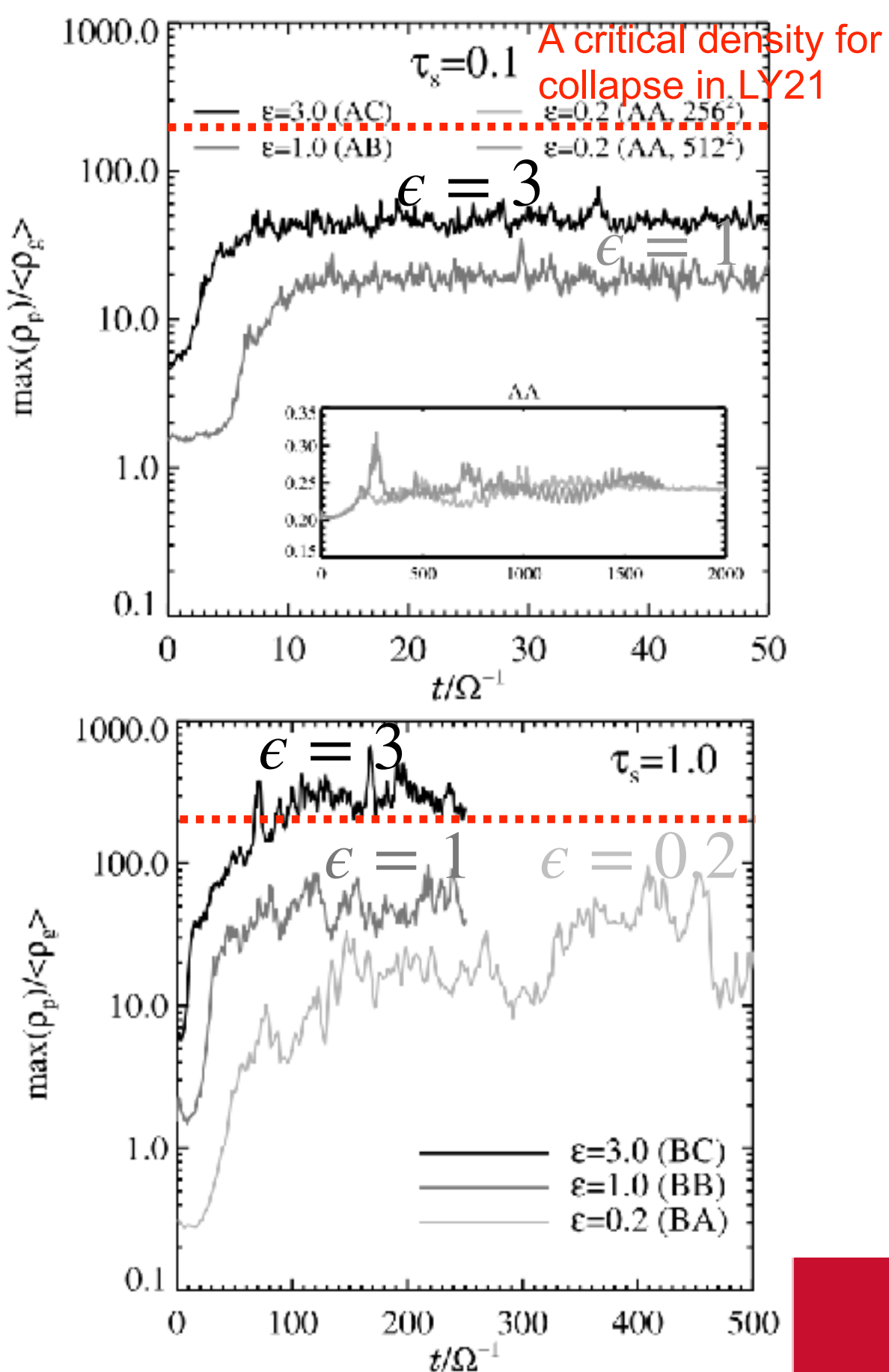
Discussion: the saturated state and strong clumping



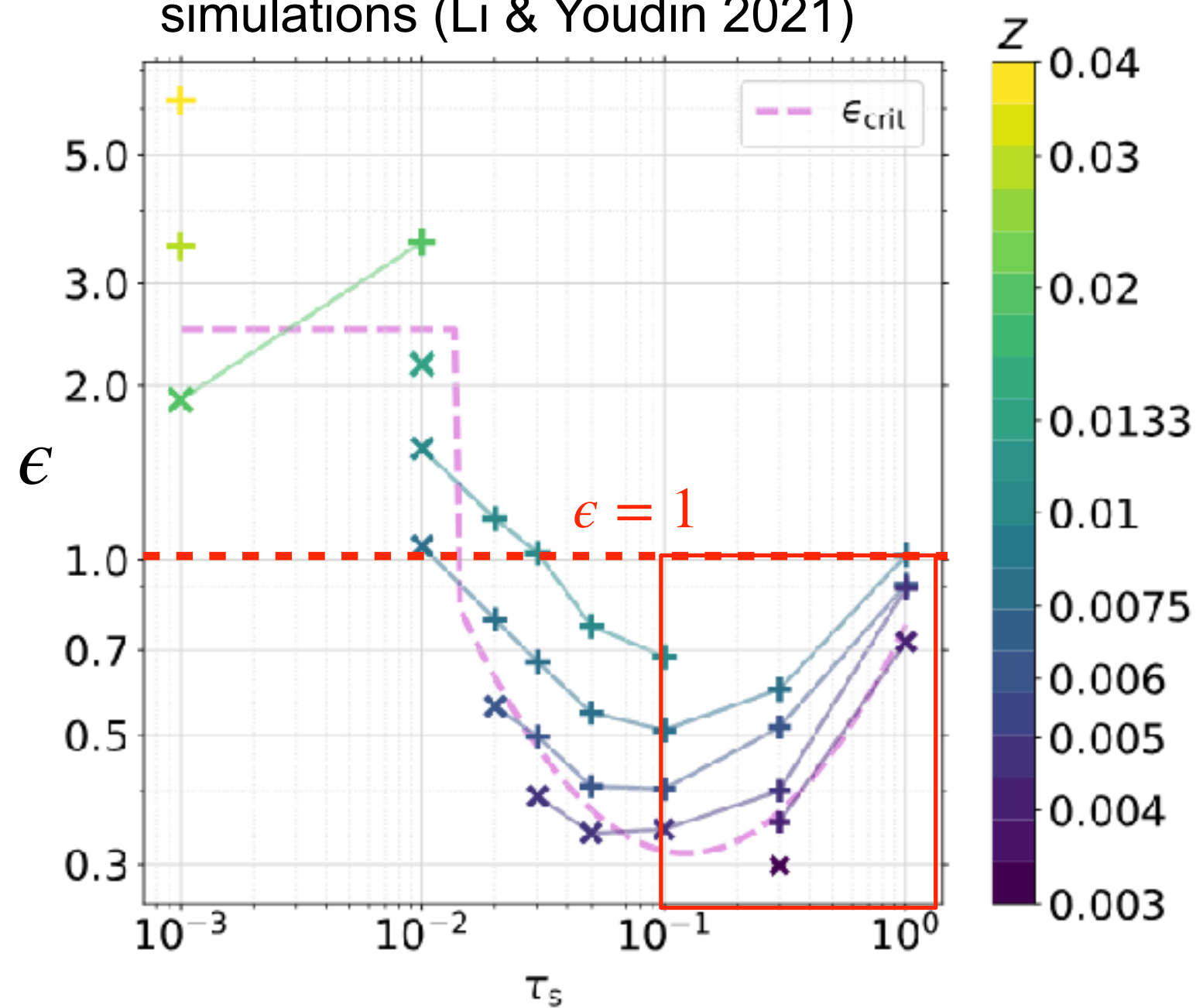
The same Z/Π value results in similar dynamics of dust and gas, but only one run shows strong clumping.

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Unstratified simulations (Johansen & Youdin 2007)



Critical ϵ value for collapse in stratified simulations (Li & Youdin 2021)



The critical ϵ value appears to be far lower in stratified cases-> what's the role of vertical gravity in triggering strong clumping?

Summary

- > The SI-driven dynamics examined for a range of pressure gradients
- > **Higher velocity fluctuations away from the midplane** before the onset of the SI.
- > **Remarkably similar dust-gas dynamics** between unstratified and stratified simulations
- > How does the saturated state determine strong clumping?
- > **Smaller critical ϵ values** in stratified simulations: What is the role of vertical gravity in triggering strong clumping?